Eunhan Ka

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Education

Purdue University *United States*
Ph.D., Lyles School of Civil and Construction Engineering *Dec. 2025*

- **Dissertation: Physics-Informed Neural Networks for Secure Connected and Autonomous Traffic Modeling**
- Committee: Satish V. Ukkusuri (Chair), Ludovic Leclercq, Yiheng Feng, Z. Berkay Celik

Seoul National University *South Korea*
M.S., Department of Civil and Environmental Engineering *Feb. 2018*

Seoul National University *South Korea*
B.S., Department of Civil and Environmental Engineering *Feb. 2016*

- Completed the ABEEK-accredited Civil and Environmental Engineering program (Engineering Accreditation Commission; Washington Accord).

Research Interests

Modeling of network traffic dynamics, resilience, and large-scale traffic state estimation, using physics-informed and data-driven methods that integrate behavioral models, game theory, and optimization across machine learning and traffic engineering.

- **Themes**: Network Traffic Dynamics, Large-Scale Traffic State Estimation, Network Resilience and Reliability, Connected and Automated Mobility, Human Behavior and Safety, Transportation Cybersecurity.
- **Methods**: Physics-Informed Learning, Graph & Network Learning, Game-Theoretic Modeling, Symbolic Learning and Scientific Discovery, Optimization and Simulation, Spatiotemporal Data Analytics

Selected Publications & Awards

Selected Publications

1. **Ka, E.** and Ukkusuri, S. V. (2026). Resilience of Traffic Networks to Route Guidance Attacks: The Role of Driver Behaviour Heterogeneity. *Transportmetrica B: Transport Dynamics*, 14, 1.
2. **Ka, E.**, Xue, J., Leclercq, L., and Ukkusuri, S. V. (2024). A Physics-Informed Machine Learning for Estimating Traffic State with a Generalized Bathtub Model in Large-scale Urban Networks. ***Transportation Research Part C: Emerging Technologies***, 164, 104661.
3. **Ka, E.** and Ukkusuri, S. V. (2025). Route Guidance Attacks in Cyber Transportation Networks: A User-Centered Study of Behavioral Sensitivity. ***Transportation Research Part F: Traffic Psychology and Behaviour***, 115, 103354.
4. Xue, J., **Ka, E.**, Feng, Y., and Ukkusuri, S. V. (2024). Network Macroscopic Fundamental Diagram-Informed Graph Learning for Traffic State Imputation. ***Transportation Research Part B: Methodological***, 189, 102996.
5. **Ka, E.**, Sharma, S., and Ukkusuri, S. V. (2022). Leveraging Location-Based Data for Assessing Network-Level Traffic Impact of Lane Management: A Case Study of Alex Fraser Bridge. *Journal of Transportation Engineering, Part A: Systems*, 148(12), 04022105. **[2022 Editor's Choice Collections]**

Selected Awards & Research Support

1. **2025 ITE Great Lakes District Student Paper Award** (First Place in regional student paper competition), *Great Lakes District of ITE (GLITE)*. Jun. 2025
2. **Honorable Mention, 2025 Clifford Spiegelman Student Paper Competition**, *The Transportation Statistics Interest Group of the American Statistical Association*. Jan. 2025
3. **ChatGPT for Academic Researchers Program, Approved Participant**, *OpenAI*. Aug. 2026
4. **Google Cloud Research Credits Program, Award Recipient**, *Google Cloud*. Aug. 2025 & Jun. 2026

Research Experience

Postdoctoral Researcher *Jan. 2026 - Present*

Purdue University, United States

- Develop physics-informed and data-assimilation methods for **transportation network state estimation, reliability assessment, and decision support** under sparse and disrupted observations.
- Build a **Resilient Mobility Digital Twin** to estimate network conditions, quantify uncertainty, and evaluate disruption propagation and recovery under climate, infrastructure, and cyber scenarios.
- Develop **behavior-aware models of connected-mobility disruptions** (e.g., route-guidance manipulation in navigation services) to evaluate network performance, operational risk, and mitigation strategies.
- Build **scalable simulation and graph-based analytical tools** for reproducible transportation-system evaluation and proposal development.

Graduate Research Assistant *Aug. 2020 - Dec. 2025*

Purdue University, United States

- Developed physics-informed and graph-based methods for **large-scale traffic-state estimation, network operations, and resilience under sparse or disrupted data**, resulting in publications in **Transportation Research Part B, Part C, and Part F**. *(Sponsor: USDOT University Transportation Center-TraCR)*
- Conducted a statewide workforce needs assessment to identify skill gaps and designed training strategies to improve engagement and performance (presented at **Purdue Road School, Mar. 2025**). *(Sponsor: INDOT)*
- Analyzed network-level impacts of a movable median barrier on the Alex Fraser Bridge using mobile location data (Journal of Transportation Engineering, Part A, 2022; **Editor's Choice**). *(Sponsor: Lindsay Corporation)*
- Led proposal development and served as the **primary technical lead throughout project execution**, coordinating analytical work, project milestones, and deliverable preparation under the supervision of Dr. Satish V. Ukkusuri (Principal Investigator). **Selected Sponsored Projects:**
  1. Attracting and Retaining the Transportation Workforce and At Risk Targeted Areas (*Sponsor: INDOT; total project award: $225,000; Oct. 2024–Sep. 2026*).
  2. A Multi-Resolution Simulation Platform for Transportation System Security Testing and Evaluation (*Sponsor: USDOT UTC-TraCR; total project budget: $340,000 including cost share; Jan.–Dec. 2024*).
  3. Training Gap Analysis for INDOT Workforce (*Sponsor: INDOT; total project award: $213,500; Oct. 2023–Oct. 2025*).

Researcher *Mar. 2018 - Aug. 2020*

Seoul National University, South Korea

- Designed cluster-based routes for demand-responsive transport using multi-capacity vehicles to serve passengers with disabilities (presented at **99th TRBAM**). *(Sponsor: T-Money Welfare Foundation)*
- Modeled lane-changing on freeways and gap-acceptance behavior at roundabouts using real-world data and virtual reality simulations (published in **Journal of Advanced Transportation, 2018**). *(Sponsor: Ministry of Land, Infrastructure and Transport, South Korea)*

Graduate Research Assistant *Mar. 2016 - Feb. 2018*

Seoul National University, South Korea

- Designed a real-time vehicle relocation strategy and an event-based simulation for one-way car-sharing services (presented at **multiple conferences, 2016-2018**). *(Sponsor: National Research Foundation of Korea)*
- Developed a methodology to evaluate driver behavior using surrogate safety measures (published in **Journal of Advanced Transportation, 2020**). *(Sponsor: Ministry of Land, Infrastructure and Transport, South Korea)*

Teaching & Mentoring Experience

Research Mentor *Fall 2024 - Present*

Lyles School of Civil and Construction Engineering, Purdue University

- **Graduate research mentoring (Jun. 2026–Present):** Mentor two Ph.D. students in transportation systems research; each student submitted a paper to the 2027 Transportation Research Board Annual Meeting.
- **Undergraduate research mentoring (Fall 2024–Spring 2025):** Mentored two undergraduate researchers in transportation cybersecurity; their work resulted in a poster presentation at the 2025 CERIAS Cybersecurity Symposium.

Teaching Assistant *Spring 2021 - Fall 2024*

LYLES SCHOOL OF CIVIL AND CONSTRUCTION ENGINEERING, PURDUE UNIVERSITY

- CE 661: Algorithms in Transportation *(Graduate; co-developed new course curriculum)* *Fall 2024*
- CE 597: Network Models for Connected and Autonomous Vehicles (Graduate) *Fall 2021 - Fall 2023*
- CE 597: Smart Logistics (Graduate) *Fall 2021, Fall 2024*
- CE 597: Data Science for Smart Cities (Graduate) *(Developed online course on **edX**)* *Spring 2021, Fall 2023*

Lecturer *Fall 2019*

DEPT. OF CIVIL AND ENVIRONMENTAL ENGINEERING, SEOUL NATIONAL UNIVERSITY

- Introduction to Civil and Environmental Engineering (Undergraduate)

Teaching Assistant *Fall 2016 - Fall 2017*

DEPT. OF CIVIL AND ENVIRONMENTAL ENGINEERING, SEOUL NATIONAL UNIVERSITY

- Sustainable Transportation Systems (Graduate) *Fall 2017*
- Advanced Transportation Operation (Graduate) *Spring 2017*
- Leadership for Civil Engineers (Undergraduate) *Spring 2017*
- Introduction to Transportation Engineering (Undergraduate) *Fall 2016*

Publications

MANUSCRIPTS UNDER REVIEW AND REVISION

1. **Ka, E.**, Leclercq, L., and Ukkusuri, S. V. (Under Major Revision). Adaptive Domain Decomposition Physics-Informed Neural Networks for Traffic State Estimation with Sparse Sensor Data. *Transportation Research Part C: Emerging Technologies*.
2. **Ka, E.**, Mittal, S., Ali, A., Liu, L., and Ukkusuri, S. V. (Under Review). Perceived regulatory calibration and cybersecurity workforce capacity in U.S. transportation: Exploratory survey evidence. *Transport Policy*.
3. **Ka, E.** and Ukkusuri, S. V. (Under Review). Day-to-Day Traffic Network Modeling under Route-Guidance Misinformation: Endogenous Trust and Resilience in CAV Environments. *IEEE Transactions on Intelligent Transportation Systems*.
4. **Ka, E.**, Yeke, D., Celik, Z. B., and Ukkusuri, S. V. (Under Review). Fake Rush Hour: Exploring Bounded Rational Route Guidance Attacks on Urban Navigation Systems. *ACM Journal on Autonomous Transportation Systems*.
5. Yang, Z., **Ka, E.**, Cai, Z., and Ukkusuri, S. V. (Under Review). Context-Matched Scenario Families with Controlled Criticality for Autonomous Driving Testing. *Transportation Research Record*.
6. **Ka, E.**, Li, H., and Ukkusuri, S. V. (Under Review). Finite-Grid Used-Path Regret Certificates for Restricted Static Traffic Assignment. *Transportation Research Record*.

REFEREED JOURNAL PUBLICATIONS

1. **Ka, E.**, Mittal, S., and Ukkusuri, S. V. (Accepted). Career-Stage-Specific Training Priorities for State DOT Workforce Development: An Integrated Regression and Importance–Performance Analysis. *Transportation Research Record: Journal of the Transportation Research Board*.
2. **Ka, E.** and Ukkusuri, S. V. (2026). Resilience of Traffic Networks to Route Guidance Attacks: The Role of Driver Behaviour Heterogeneity. *Transportmetrica B: Transport Dynamics*, 14, 1.
3. Ali, A., Liu, L., Ojeme, B., Anadozie, C., Agyemang, F., Ukkusuri, S. V., **Ka, E.**, and Mittal, S. (2026). Strengthening the Transportation Cybersecurity Workforce: A Mixed-Methods Analysis of Workforce Development, Organizational Preparedness and Cross-Sector Collaboration. *Future Transportation*. 6(4), 140
4. **Ka, E.** and Ukkusuri, S. V. (2025). Route Guidance Attacks in Cyber Transportation Networks: A User-Centered Study of Behavioral Sensitivity. *Transportation Research Part F: Traffic Psychology and Behaviour*, 115, 103354.
5. Ukkusuri, S. V.*, Hamim, O. F.*, Lei, Z.*, **Ka, E.***, Salek, M. S., Chowdhury, M., Amini, M. H., Cardenas, A., and Thuraisingham B. (2025). Cybersecurity for Next-Generation Road Transportation: A Review. *ACM Journal on Autonomous Transportation Systems*. (*Contributed equally to this research).
6. **Ka, E.**, Xue, J., Leclercq, L., and Ukkusuri, S. V. (2024). A Physics-Informed Machine Learning for Estimating Traffic State with a Generalized Bathtub Model in Large-scale Urban Networks. *Transportation Research Part C: Emerging Technologies*, 164, 104661.

7. Xue, J., **Ka, E.**, Feng, Y., and Ukkusuri, S. V. (2024). Network Macroscopic Fundamental Diagram-Informed Graph Learning for Traffic State Imputation. *Transportation Research Part B: Methodological*, 189, 102996.
8. **Ka, E.**, Sharma, S., and Ukkusuri, S. V. (2022). Leveraging Location-Based Data for Assessing Network-Level Traffic Impact of Lane Management: A Case Study of Alex Fraser Bridge. *Journal of Transportation Engineering, Part A: Systems*, 148(12), 04022105. **[2022 Editor's Choice Collections]**
9. **Ka, E.**, Kim, D. G., Hong, J., and Lee, C. (2020). Implementing Surrogate Safety Measures in Driving Simulator and Evaluating the Safety Effects of Simulator-Based Training on Risky Driving Behaviors. *Journal of Advanced Transportation*, 2020(1), 7525721.
10. Lee, D., Hwang, S., **Ka, E.**, and Lee, C. (2018). Evaluation of the Rain Effects on Gap Acceptance Behavior at Roundabouts by a Logit Model. *Journal of Advanced Transportation*, 2018(1), 2726732.

Conference Papers and Presentations

1. **Ka, E.**, Yeke, D., Celik, Z. B., and Ukkusuri, S. V. (2026). MIRAGE: Detecting Fake Emergency Electronic Brake Light Attacks in V2X Networks via Event-Gated Behavioral Analysis. *4th USENIX Symposium on Vehicle Security and Privacy (VehicleSec'26)*. Baltimore, MD, United States
2. Ortiz Barbosa, D., Abdisho, E. B., Chang, O., Burbano, L., Lei, Z., **Ka, E.**, Ukkusuri, S. V., Gilpin, L. H., and Cardenas, A. A. (2026). Beyond Attacker-Defender Ratios in Counter-Swarm Defense. *Dynamic Data Driven Applications Systems (DDDAS) 6th International Conference*. New Brunswick, NJ, United States
3. **Ka, E.** and Ukkusuri, S. V. (2025). Analytical Framework for Network-Level Traffic Flow under Route Guidance Attacks: An Extension of the Generalized Bathtub Model. *104th Annual Meeting of the Transportation Research Board*, Washington, D.C., United States.
4. **Ka, E.** and Ukkusuri, S. V. (2025). Comparative Analysis of Sampling Methods in Physics-Informed Neural Networks for Traffic State Estimation in Large-Scale Road Networks. *104th Annual Meeting of the Transportation Research Board*, Washington, D.C., United States.
5. **Ka, E.**, and Ukkusuri, S. V. (2024). Impact of Cyber Attacks on Traffic State Estimation for Connected and Autonomous Vehicles Systems. *The Inaugural USDOT Future of Transportation (FoT) Summit*, Washington, D.C., United States.
6. Xue, J., **Ka, E.**, and Ukkusuri, S. V. (2024). Network Macroscopic Fundamental Diagram-Informed Graph Learning for Traffic State Imputation. *ISTTT25: 25th International Symposium on Transportation and Traffic Theory*, Ann Arbor, MI, United States.
7. **Ka, E.**, Xue, J., and Ukkusuri, S. V. (2024). PIDL-PedFlow: A Physics-Informed Deep Learning Approach for Macroscopic Continuum Pedestrian Flow Modelling. *103rd Annual Meeting of the Transportation Research Board*, Washington, D.C., United States.
8. Xue, J., **Ka, E.**, Mondal, W. U., and Ukkusuri, S. V. (2024). Generating Network-Level Dynamic Traffic Equations Using Symbolic Regression. *103rd Annual Meeting of the Transportation Research Board*, Washington, D.C., United States.
9. Sharma, S., and **Ka, E.** (2024). Leveraging Location-Based Data for Assessing Network Level Traffic Impact of Lane Management: A Case Study of Alex Fraser Bridge. *103rd Annual Meeting of the Transportation Research Board*, Washington, D.C., United States.
10. **Ka, E.**, and Ukkusuri, S. V. (2023). Dynamic Routing Games for Connected and Autonomous Vehicles with Traffic Congestion: A Mean Field Game Approach. *2023 INFORMS Annual Meeting*, Phoenix, AZ, United States.
11. **Ka, E.**, Ka, D., Jung, Y., and Lee, C. (2020). A Cluster-Based Route Design of Multi-Capacity Vehicle in Large-Scale Demand Responsive Transport Service for the Disabled. *99th Annual Meeting of the Transportation Research Board*, Washington, D.C., United States.
12. **Ka, E.**, Hong, D., Na, Y., and Lee, C. (2018). Analysis of Status in DRT Service for the Disabled in Seoul and Comparison Domestic and Foreign Cases. *International Conference for Road Engineers*, Jeju, South Korea.
13. Hong, D., **Ka, E.**, Ha, S., and Lee, C. (2018). Selection of Appropriate Hyperparameter for Waiting Time Prediction Model for Demand Responsive Transport for the Disabled in Seoul Using Long Short-Term Memory (LSTM) Network. *78th Korean Society of Transportation Conference*, Wonju, South Korea. **[Outstanding Paper Award]**.
14. **Ka, E.**, Kim, S., Hong, J., and Lee, C. (2017). A Preliminary Study of Comparison with Lane Changing Model Parameters in Merging Area between Normal and Raining Conditions. *12th International Conference of Eastern Asia's Society for Transportation Studies (EASTS)*, Ho Chi Minh, Vietnam.

15. **Ka, E.**, Kim, S., Woo, D., and Lee, C. (2016). An Estimation of Critical Gap for Gap Acceptance Model Applied to Lane Change of Surrounding Vehicles in Driving Simulator. *2016 Fall Korea Institute of ITS Conference*, Jeju, South Korea.
16. Lee, S., Lee, H., Lee, J., **Ka, E.**, and Lee, C. (2016). Analysis of Traffic Flow Impacts of Highway Traffic Accidents Using Survival Analysis. *74th Korean Society of Transportation Conference*, Jeju, South Korea. **[Outstanding Paper Award]**.

Book Chapters and Technical Reports

1. **Ka, E.** and Ukkusuri, S. V. (2026). Misdirected Guidance and Urban Traffic Network Resilience. In *Advances in Transportation Cybersecurity and Resiliency*. World Scientific.
2. Hamim, O. F., Lei, Z., **Ka, E.**, and Ukkusuri, S. V. (2026). Understanding Cyber Threats in Cyber-Physical Mobility Systems. In *Advances in Transportation Cybersecurity and Resiliency*. World Scientific.
3. Verma, R., Luo, H., Deodhar, S., **Ka, E.**, Chahine, R., Natu, P., Malhotra, H., Polisetty, V., Thakkar, D. J., Ukkusuri, S. V., Cai, H., Dunlop, S. R., Iyer, A. V., and Gkritza, K. (2023). *Forecasting Shifts in Hoosiers' Travel Demand and Behavior* (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2023/28). Purdue University.

Patents and Intellectual Property

1. Ukkusuri, S. V. and **Ka, E.** (2026). System and Method for Trusted Route Guidance with Anomaly Detection. *U.S. Provisional Patent Application No. 63/989,950, filed Feb. 24, 2026. Purdue Research Foundation. Patent pending*.

Awards, Fellowships & Research Support

ChatGPT for Academic Researchers Program, Approved Participant *Aug. 2026*
OpenAI

Google Cloud Research Credits Program, Award Recipient *Jun. 2026*
Google Cloud

Google Cloud Research Credits Program, Award Recipient *Aug. 2025*
Google Cloud

2025 ITE Great Lakes District Student Paper Award & GLITE Endowment Fund *Jun. 2025*
Great Lakes District of Institute of Transportation Engineers (GLITE)

Honorable Mention, 2025 Clifford Spiegelman Student Paper Competition *Jan. 2025*
The Transportation Statistics Interest Group of the American Statistical Association

2025 KOTAA Travel Grant Award *Jan. 2025*
Korean Transportation Association in America (KOTAA)

Student Paper Award *Dec. 2024*
ITE Indiana

Kinnier Graduate Scholarship *Oct. 2024*
Lyles School of Civil and Construction Engineering, Purdue University

STV Civil Engineering Graduate Assistantship Endowment *Oct. 2024*
Lyles School of Civil and Construction Engineering, Purdue University

2024 KOTAA Travel Grant Award *Jan. 2024*
Korean Transportation Association in America (KOTAA)

Crooks Travel Scholarship *Sep. 2023*
Lyles School of Civil Engineering, Purdue University

Outstanding Paper Award *Mar. 2018*
78th Conference of Korean Society of Transportation, Korean Society of Transportation

Outstanding Paper Award *Feb. 2016*
74th Conference of Korean Society of Transportation, Korean Society of Transportation

Selected Presentations and Invited Talks

1. Physics-Informed AI for Transportation Systems. *CGN 6425: Applied Data Science in Civil and Environmental Engineering, University of Florida*. April 2, 2026.

2. Needs and Gap Assessment of INDOT Workforce Development. *2025 Purdue Road School Transportation Conference and Expo*. March 18, 2025.
3. A Physics-Informed Machine Learning for Generalized Bathtub Model in Large-Scale Urban Networks. *TRB ACP50 Committee on Traffic Flow Theory and Characteristics General Webinar Series*. March 17, 2023.

Academic Service and Professional Activities

Academic Service

- **Journal Referee**: IEEE Transactions on Intelligent Transportation Systems; Transportation Research Part A: Policy and Practice; Transportation Research Part B: Methodological; Transportation Research Part C: Emerging Technologies; Transportation Research Part F: Traffic Psychology and Behaviour; Transportation Research Interdisciplinary Perspectives; Transportation; Engineering Applications of Artificial Intelligence; Scientific Reports; Electric Power Systems Research; The Journal of Supercomputing; Journal of Cloud Computing; Transportation Research Record: Journal of the Transportation Research Board; Journal of Transportation Engineering, Part A: Systems; Data Science for Transportation
- **Conference Referee**: Transportation Research Board; IEEE Intelligent Transportation Systems Conference (ITSC); ISTTT25 (Co-reviewer); ISTTT26 (Co-reviewer)

Professional Activities

- **Association for Computing Machinery (ACM)**: Professional Member (Aug. 2025 - Current)
- **American Society of Civil Engineers (ASCE)**: Associate Member (May 2025 - Current); Student Member (Sep. 2022 - Apr. 2025); **AI Committee**: Corresponding Member (Aug. 2025 - Current)
- **Institute of Electrical and Electronics Engineers (IEEE)**: Member (May 2025 - Current)
- **Institute for Operations Research and the Management Sciences (INFORMS)**: Member (May 2023 - Current)
- **Institute of Transportation Engineers (ITE) Purdue Student Chapter**: Communications Director (Aug. 2024 - July 2025); Member (Aug. 2020 - Current)

Technical Skills

- **Transportation Modeling and Simulation** : SUMO, PTV Vissim, TransModeler, TransCAD
- **Geospatial and Statistical Analysis** : ArcGIS, SAS, R, GeoPandas, SPSS, NLOGIT
- **Programming, Machine Learning, and Optimization** : Python, PyTorch, TensorFlow, Keras, PySpark, Java, MATLAB, CPLEX
- **Reproducible Research and Documentation** : Git, Jupyter, LaTeX